

RESTLESS EARTH

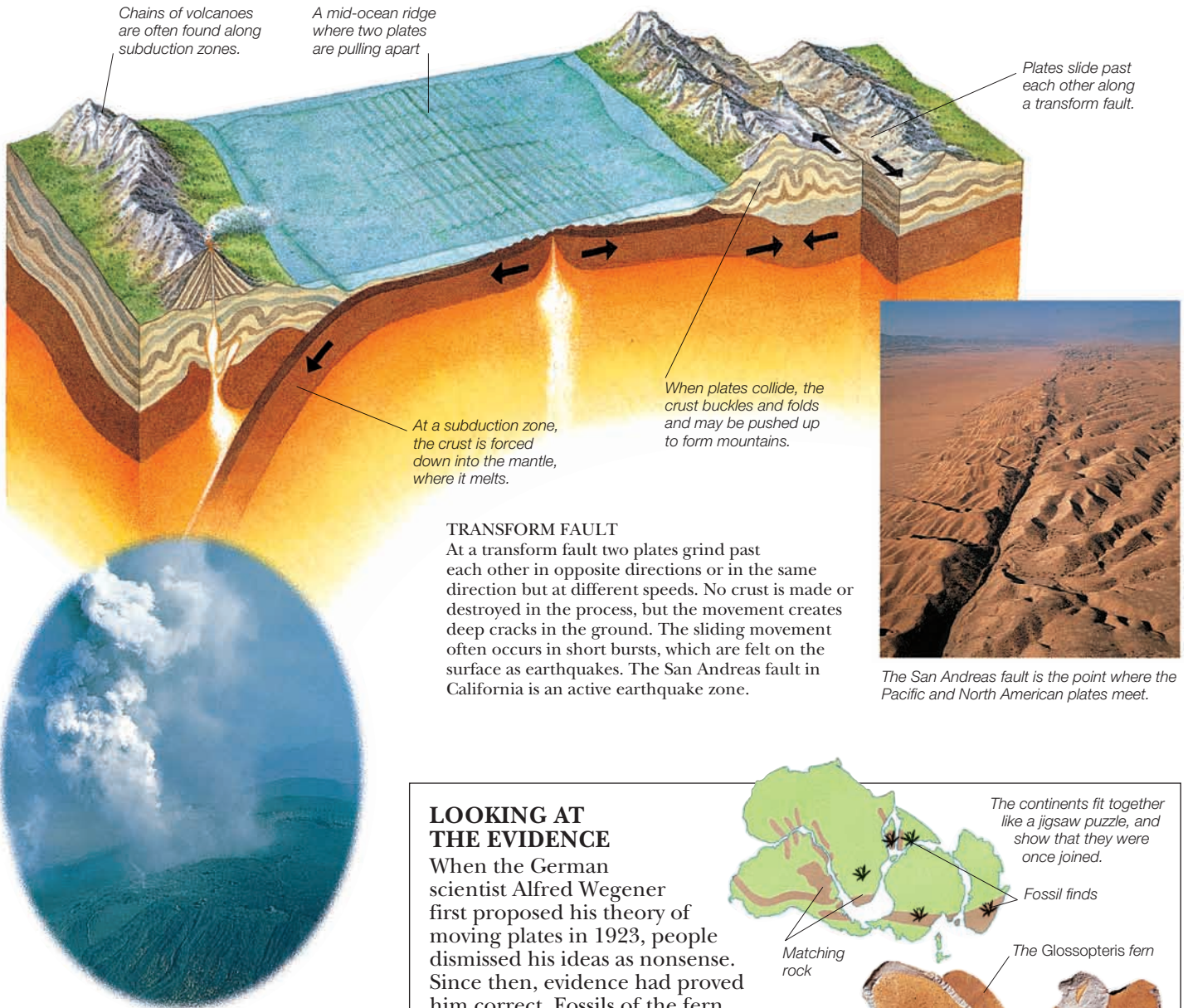
Because the Earth appears to stand still, it is difficult to imagine that the crust is moving. In fact, its plates move in three main ways – as spreading ridges, subduction zones, and transform faults, all shown on the artwork below. It is possible to see the effect this activity has had on the landscape. The Rocky Mountains in North America were formed when two plates collided, while the Great Rift Valley in Africa is the result of plates pulling apart. Volcanoes and earthquakes are also dramatic reminders that the plates are moving.



At Thingvellir, Iceland, the spreading ridge between the North American and Eurasian plates appears as a long gash in the landscape.

SPREADING RIDGES

A spreading ridge occurs where two plates start to pull apart and molten rocks from the Earth's mantle well up to fill the gap. If this happens along the ocean floor, it creates an underwater mountain chain such as the Mid-Atlantic Ridge. Sometimes the peaks of these mountains break the surface as volcanic islands, as happened with Iceland. When a spreading ridge occurs on land, it creates a steep-sided rift valley.



TRANSFORM FAULT

At a transform fault two plates grind past each other in opposite directions or in the same direction but at different speeds. No crust is made or destroyed in the process, but the movement creates deep cracks in the ground. The sliding movement often occurs in short bursts, which are felt on the surface as earthquakes. The San Andreas fault in California is an active earthquake zone.

The San Andreas fault is the point where the Pacific and North American plates meet.

SUBDUCTION ZONE

When two plates meet, the edge of one can be pushed down (subducted) under the other and into the mantle below. The rocks from the crust melt in the mantle. Often these molten rocks force their way to the surface as a volcano. The many volcanoes around the edge of the Pacific plate, such as Mt. Mihara, Japan, were formed this way. Sometimes when plates collide, rocks are forced up, forming great mountain ranges.

LOOKING AT THE EVIDENCE

When the German scientist Alfred Wegener first proposed his theory of moving plates in 1923, people dismissed his ideas as nonsense. Since then, evidence had proved him correct. Fossils of the fern *Glossopteris* for example, have been found in rocks as far apart as India, Australia, and Africa. All these places were once joined together as Gondwanaland. Further proof comes from matching types of rock that have been found in Australia, Antarctica, and South America.

